

SoccerNet 2026 Player-Centric Ball-Action Spotting: Retraining and Post-Processing Extensions to the FOOTPASS Baselines

Parthsarathi Rawat
GameChanger by Dick’s Sporting Goods
sarathi.rawat@gc.com

Abstract

*We describe our system for the SoccerNet 2026 Player-Centric Ball-Action Spotting Challenge, which requires predicting who performs which action and when, across eight classes in broadcast soccer. Building on the three FOOTPASS baselines [1] (TAAD, TAAD+GNN, and TAAD+DST), we contribute four extensions: (1) gradient checkpointing to enable full-backbone fine-tuning on a single GPU; (2) fusion of GNN logits into the DST encoder; combining graph-based tactical context with per-player visual features; (3) square-root frequency class weighting to address the 213:1 pass-to-tackle imbalance in the training data; and (4) a post-processing pipeline comprising per-class logit gating, temporal frame refinement, jersey re-assignment, and a two-model ensemble. Our system achieves **0.548 Macro F1** on the test set and **0.446** on the challenge set (server evaluation).*

1. FOOTPASS Baseline System

The challenge organisers provide three reference model architectures and training code [1]; we trained all models from scratch. The primary metric is **Macro F1**: the unweighted average of per-class F1 across all eight action classes, giving equal importance to tackle (24 ground-truth events in the challenge set) and pass (2 744 events). The three baselines form a cascade: TAAD produces noisy per-player action logits from video; TAAD+GNN enriches those logits with player-graph context; TAAD+DST reads the full sequence of logit tracks and decodes a clean event list.

TAAD [2, 3]. An X3D-S backbone [4] shared across $M=26$ tracked players extracts multi-scale video features, which are fused via lateral upsampling into a $(B, 192, T, 44, 80)$ tensor. Per-player crops are extracted with RoIAlign [5] (4×2 grid, scale 0.125); a temporal Conv1d($192 \rightarrow 512, k=3$) and linear head yield 9-class logits (background plus eight actions) per player per frame.

TAAD+GNN [2]. An EdgeConv [6] branch models interactions among players. Each player-frame node carries a

69-dimensional feature (pitch coordinates, velocities, one-hot role encoding, jersey number, and team side) concatenated with a 64-dimensional visual projection from the same RoIAlign crop; each player is connected to its six nearest spatial neighbours. Three EdgeConv layers (hidden dim 128, max-aggregation) interleaved with temporal Conv1d($k=5$) blocks propagate spatial and temporal context, yielding a second set of 9-class per-player logits that capture who is near the ball and what role they play.

TAAD+DST [7]. Rather than thresholding TAAD logits independently per frame, the Denoising Sequence Transducer (DST) [8] reads the entire sequence of per-player logit tracks as a source and autoregressively decodes a de-duplicated list of (action, player, frame) events. This allows the model to suppress spurious detections by attending to context across the full 25-second window. The default configuration uses 364-dimensional source tokens, hidden dimension 512, 6 encoder and 6 decoder layers, and 8 attention heads over a 750-frame context window. The model is trained with cross-entropy loss and label smoothing 0.05. At a 15% global confidence threshold this baseline achieves 0.493 Macro F1 on the test set.

2. Our Extensions

The four extensions address training capacity, feature richness, class imbalance, and inference quality, in that order.

2.1. Gradient Checkpointing for Full Backbone Fine-Tuning

The baseline TAAD training script freezes the X3D backbone after the initial epochs because unfreezing at batch size 14 exceeds the 22 GB GPU memory limit. We add gradient checkpointing on X3D blocks 0–4, enabling full fine-tuning at batch size 6. Training runs 20 epochs with the backbone frozen for epochs 1–2 then unfrozen, using AdamW (lr 5×10^{-5} backbone, 10^{-3} head, 50-step warm-up).